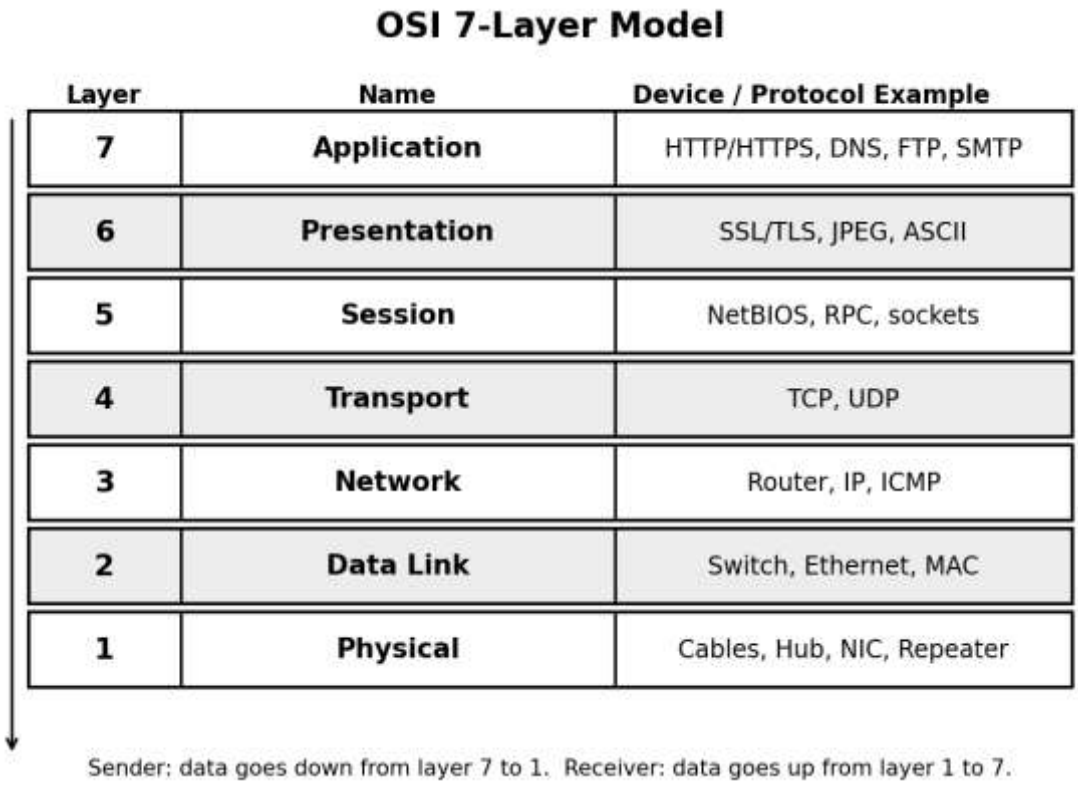
OSI Layer

Assignment Answers

Q1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).



The OSI (Open Systems Interconnection) model divides network communication into 7 layers. Each layer has its own job and talks to the layer above and below it. The table below explains each layer.

No.	Layer	Function	Device / Protocol Example
7	Application	Gives network services directly to the user's apps, such as browsing, email and file transfer.	HTTP/HTTPS (web browser), DNS, FTP, SMTP
6	Presentation	Translates, encrypts and compresses data so the receiver can understand it.	SSL/TLS (encryption), JPEG, ASCII
5	Session	Starts, manages and ends the connection (session) between two devices.	NetBIOS, RPC, sockets
4	Transport	Breaks data into segments, controls flow and gives reliable (TCP) or fast (UDP) delivery.	TCP, UDP, firewall (port filtering)
3	Network	Adds IP addresses and finds the best path (routing) for packets between networks.	Router, IP, ICMP (ping)
2	Data Link	Puts data into frames, uses MAC addresses and detects errors between devices on the same network.	Switch, Bridge, Ethernet, MAC address
1	Physical	Sends raw bits as electrical, light or radio signals over the medium.	Cables, Hub, Repeater, NIC, WiFi signal

Q2Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames. Write your answers as 'Scenario: Layer Name (Layer Number).

a) Encrypting data before sending: Presentation Layer (Layer 6). This layer handles encryption, decryption and data formatting (for example SSL/TLS) before the data is passed down.

b) Routing packets across networks: Network Layer (Layer 3). Routers use IP addresses at this layer to find the best path for packets to reach another network.

c) Converting data to electrical signals: Physical Layer (Layer 1). It converts bits (0s and 1s) into electrical signals, light pulses or radio waves and sends them over the cable or air.

d) Opening a web page in your browser: Application Layer (Layer 7). The browser uses protocols like HTTP/HTTPS at this layer to request and show the web page.

e) Detecting and correcting errors in frames: Data Link Layer (Layer 2). It uses frames and error checking (such as the CRC/FCS check) to find damaged frames so they can be dropped or sent again.

Answers in the required format

- Encrypting data before sending: Presentation (6)
- Routing packets across networks: Network (3)
- Converting data to electrical signals: Physical (1)
- Opening a web page in your browser: Application (7)
- Detecting and correcting errors in frames: Data Link (2)

Q3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

Scenario: I type "Hello" in WhatsApp and press Send. The message passes down the 7 layers on my phone, travels over the network, and passes up the 7 layers on the receiver's phone.

No.	Layer	What happens in WhatsApp
7	Application	The WhatsApp app itself takes my message and uses an application protocol over HTTPS to hand it to the network. This is the layer I interact with.
6	Presentation	The message is encrypted (end-to-end encryption) and the text, photo or voice note is converted and compressed into a format that can be sent.
5	Session	A session is opened between my phone and the WhatsApp server, and kept alive so messages, "typing" and "online" status can be exchanged, then closed when finished.
4	Transport	The message is broken into segments. TCP numbers them and makes sure all arrive in order, and the port number identifies the WhatsApp connection.
3	Network	IP addresses of my phone and the WhatsApp server are added to each packet, and routers decide the path across the internet.
2	Data Link	Packets are put into frames with MAC addresses, so they pass from my phone to the WiFi router (or tower) and from device to device. Errors in frames are checked.
1	Physical	The frames are changed into radio waves (WiFi/mobile data), and then into electrical or light signals on the cables to reach the server.

Receiving: The receiver's phone gets the signals at layer 1 and moves them up through layers 2, 3, 4, 5 and 6. At layer 6 the message is decrypted, and at layer 7 WhatsApp shows "Hello" in the chat.

Q4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Observation: The ping to the IP address is successful, but the website cannot be opened.

Layers that are working

Ping uses ICMP and works at the Network layer. A successful ping shows that the Physical layer (cable or WiFi is connected), the Data Link layer (frames, MAC and switch are working) and the Network layer (IP addressing and routing are correct) are all working. So the problem is not in layers 1, 2 or 3.

Layers that may be causing the problem

Application Layer (Layer 7) - most likely: Opening a website needs DNS to change the name (for example www.example.com) into an IP address. If DNS is wrong or not working, the name cannot be found even though the IP address itself can be reached. The browser, proxy settings or the website's web service (HTTP) may also be faulty.

Transport Layer (Layer 4): Ping does not use TCP or UDP ports. A web page uses TCP port 80 (HTTP) or 443 (HTTPS). A firewall or the website server may be blocking these ports, so ping succeeds while the website fails.

Presentation Layer (Layer 6) - less likely: An invalid or expired SSL/TLS certificate, or a wrong date and time on the computer, can stop an HTTPS website from opening.

Quick steps to find the exact cause

- Run nslookup or ping the website name. If the name does not resolve, the DNS (Layer 7) is the problem; change the DNS to 8.8.8.8.
- Check whether port 80 or 443 is open (firewall or antivirus). If it is blocked, the problem is in Layer 4.
- Try another browser, clear the cache and turn off the proxy or VPN.

Conclusion: Since ping works, layers 1 to 3 are fine. The fault is in the upper layers, most likely Layer 7 (DNS or web service) or Layer 4 (blocked port), and sometimes Layer 6 (certificate).